

3. Find the covariance and correlation of given data points.

a) $X = [2, 4, 5, 6, 7, 8, 9, 10]$

$Y = [55, 60, 65, 70, 75, 80, 85, 90]$

Here,

$$\text{Covariance} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

$$\text{Mean}(\bar{X}) = \frac{2+4+5+6+7+8+9+10}{8} = \frac{51}{8} = 6.375$$

$$\text{Mean}(\bar{Y}) = \frac{55+60+65+70+75+80+85+90}{8} = \frac{580}{8} = 72.5$$

Deviations

x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x}) \cdot (y - \bar{y})$
2	55	-4.375	-17.5	76.56
4	60	-2.375	-12.5	29.68
5	65	-1.375	-7.5	10.31
6	70	-0.375	-2.5	0.93
7	75	0.625	2.5	1.56
8	80	1.625	7.5	12.18
9	85	2.625	12.5	32.81
10	90	3.625	17.5	63.43

Now,

$$\text{Covariance} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1} = \frac{227.5}{8-1} = 32.5$$

$\therefore$ Positive linear relationship between x and y .

$$\text{Correlation} = \frac{\text{Cov}(x, y)}{\sigma_x \cdot \sigma_y} = 32.5$$

$$\sigma = \sum (x - \bar{x})^2$$

$$= 49875$$

$$\sigma = \sum (y - \bar{y})^2$$

$$= 1050$$

Variance of x

$$\sigma = \frac{49875}{7}$$

$$= 7125$$

Variance of $y = \frac{1050}{7}$

$$= 150$$

$$SD = \sqrt{7125}$$

$$= 84.41$$

$$SD = \sqrt{150}$$

$$= 12.24$$

Correlation $r = \frac{Cov(x, y)}{\sigma_x \cdot \sigma_y}$

$$= \frac{32.5}{84.41 \times 12.24}$$

$$= \frac{32.5}{1033.69}$$

$$= 0.03$$

$\therefore$ Positive linear relationship between a and b

$$a = [0, 50, 100, 150, 200, 250, 300, 350]$$

$$b = [60, 75, 88, 95, 92, 80, 65, 40]$$

Here

$$\text{Mean}(\bar{a}) = \frac{0+50+100+150+200+250+300+350}{8} = \frac{1400}{8} = 175$$

$$\text{Mean}(\bar{b}) = \frac{60+75+88+95+92+80+65+40}{8} = \frac{695}{8} = 86.875$$

Deviations

a	b	$a - \bar{a}$	$b - \bar{b}$	$(a - \bar{a})^2$	$(b - \bar{b})^2$	$(a - \bar{a}) \cdot (b - \bar{b})$
0	50	-175	-34.37	30625	206.64	-2515.62
50	75	-125	0.625	15625	0.39	-28.12
100	88	-75	13.025	5625	185.64	-1021.82
150	95	-25	20.02	625	425.39	-515.62
200	92	25	17.62	625	310.64	440.62
250	80	75	5.62	5625	31.64	421.82
300	65	125	-9.37	15625	87.89	-1171.87
350	90	175	-34.37	30625	1181.64	-6015.65
				105000	2429.875	-5925

$$\text{Covariance} = \frac{\sum (a - \bar{a})(b - \bar{b})}{n - 1} = \frac{-5925}{8} = -775$$

SD of a

$$\begin{aligned}
 S_a &= \sqrt{\frac{\sum (a - \bar{a})^2}{n - 1}} \\
 &= \sqrt{\frac{105000}{7}} \\
 &= \sqrt{15000} \\
 &= 122.474
 \end{aligned}$$

SD of b

$$\begin{aligned}
 S_b &= \sqrt{\frac{\sum (b - \bar{b})^2}{n - 1}} \\
 &= \sqrt{\frac{2429.875}{7}} \\
 &= \sqrt{347.125} \\
 &= 18.632
 \end{aligned}$$

$$\text{Correlation, } r = \frac{\text{Cov}(a, b)}{S_a \cdot S_b}$$

$$= \frac{-775}{122.474 \times 18.632}$$

$$= -0.39$$

$\therefore$ Negative linear relationship between a & b.

$$a = [2, 4, 5, 6, 7, 8, 9, 10]$$

$$b = [90, 85, 80, 75, 70, 65, 60, 55]$$

Here,

$$\text{Mean}(\bar{a}) = \frac{2+4+5+6+7+8+9+10}{8} = \frac{51}{8} = 6.375$$

$$\text{Mean}(\bar{b}) = \frac{90+85+80+75+70+65+60+55}{8} = \frac{580}{8} = 72.5$$

Deviations

a	b	(a - $\bar{a}$)	(b - $\bar{b}$)	(a - $\bar{a}$) ²	(b - $\bar{b}$) ²	(a - $\bar{a}$) · (b - $\bar{b}$)
2	90	-4.375	17.5	19.14	306.25	-76.62
4	85	-2.375	12.5	5.64	156.25	-29.68
5	80	-1.375	7.5	1.89	56.25	-10.31
6	75	-0.375	2.5	0.14	6.25	-0.93
7	70	0.625	-2.5	0.39	6.25	-1.56
8	65	1.625	-7.5	2.64	56.25	-12.18
9	60	2.625	-12.5	6.89	156.25	-32.81
10	55	3.62	-17.5	13.14	306.25	-63.43
				49.875	1050	-227.5

Now,

$$\text{Covariance} = \frac{\sum (a - \bar{a})(b - \bar{b})}{n-1} = \frac{-227.5}{8-1} = -32.5$$

$$S_a = \sqrt{\frac{\sum (a - \bar{a})^2}{n-1}} = \sqrt{\frac{49.875}{7}} = \sqrt{7.125} = 2.669$$

$$S_b = \sqrt{\frac{\sum (b - \bar{b})^2}{n-1}} = \sqrt{\frac{1050}{7}} = \sqrt{150} = 12.247$$

$$\text{Correlation, } r = \frac{\text{Cov}(a, b)}{S_a \cdot S_b} = \frac{-32.5}{2.669 \times 12.247} = \frac{-32.5}{32.692} = -0.99$$

∴ Negative relationship between a & b.

1. Find the z-scores for your marks in MCQ1, MCQ2 and MCQ3 treating the corresponding results as raw data.

Here

$$MCQ1 = 9$$

$$MCQ2 = 14$$

$$MCQ3 = 7$$

$$\text{Mean}(\bar{x}) = \frac{9+14+7}{3} = \frac{30}{3} = 10$$

Deviations

x	$(x - \bar{x})$	$(x - \bar{x})^2$
9	-1	1
14	4	16
7	-3	9
$\Sigma(x - \bar{x})^2$		26

$$\begin{aligned} \text{Variance} &= \frac{\Sigma(x - \bar{x})^2}{n} \\ &= \frac{26}{3} = 8.667 \end{aligned}$$

$$\begin{aligned} \text{SD} &= \sqrt{\text{variance}} \\ &= \sqrt{8.667} \\ &= 2.944 \end{aligned}$$

2 scores

$$\begin{aligned}MCQ1 &= \frac{x - \mu}{SD} \\&= \frac{9 - 10}{2.944} \\&= -0.34\end{aligned}$$

$$\begin{aligned}MCQ2 &= \frac{x - \mu}{SD} \\&= \frac{14 - 10}{2.944} \\&= 1.36\end{aligned}$$

$$\begin{aligned}MCQ3 &= \frac{x - \mu}{SD} \\&= \frac{7 - 10}{2.944} \\&= -1.02\end{aligned}$$

2. Find 40th, 60th, 80th percentile from MCQ1, MCQ2, MCQ3 result.
Given

$$MCQ1 = 9$$

$$MCQ2 = 14$$

$$MCQ3 = 7$$

Arranging data in ascending order: 7, 9, 14

$$n = 3$$

$$\text{When } p = 80\%$$

$$\text{Now, when } p = 40$$

$$\text{index} = \frac{p \cdot N}{100}$$

$$\text{index} = \frac{p \cdot N}{100}$$

$$= \frac{40 \times 3}{100}$$

$$= \frac{80 \times 3}{100}$$

$$= 2.4 \approx 3 \text{ (7)}$$

$$= 1.2 \approx 2 \text{ (9)}$$

$$\text{When } p = 60\%$$

Since, index 2 is already occupied by 40th,

$$\text{index} = \frac{p \cdot N}{100}$$

$$\text{for } 60\%, \quad p = \frac{\text{data}(i) + \text{data}(i+1)}{2}$$

$$= \frac{60 \times 3}{100}$$

$$= \frac{40 + 60}{2}$$

$$= 1.8 \approx 2$$

$$= 50$$